

★ WEEK 4 · PART 1 · PRACTICE

The ARIMA Model, Worked by Hand

Difference a Price Into Stationarity, Then Forecast the Trend

The 'I' in ARIMA, the order d , and integrating back

Part 1's core operation, with the arithmetic shown. We difference a trending price series until it's stationary, read off the order of integration, then forecast the level by integrating the differences back - watching the stochastic trend continue rather than revert.

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PART 1 · APPLICATIONS

One Difference Makes It Stationary

Case A: difference a price series and confirm the changes are stationary while the level trends - the order of integration is one. Case B: identify the ARIMA order, forecast the differenced series, and integrate back so the trend continues. Then a Claude Code lab.

What Part 1 Told Us

Now we apply it - with numbers

1

The 'I' is integration - differencing

ARIMA extends ARMA to trending series by differencing them first. Each difference turns a level into a change.

2

The order d counts the differences

How many differences make the series stationary. A random walk needs one, so it's ARIMA(0,1,0).

3

Forecasting integrates the differences back

Model the stationary differences, then cumulate them - so the stochastic trend continues rather than reverting.

Everything here is a calculator exercise: one difference, one order, one integrated forecast.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a trending price series.
- Difference it, or forecast and integrate back.
- Then read the trend.

What you should watch

- How differencing removes the trend.
- How the order d is decided.
- How the integrated forecast keeps trending.

Differencing to Stationarity

A trend removed in one step

A price series wanders upward - non-stationary, with a stochastic trend. ARIMA's first move is to difference it: replace each level with its change from the day before. The differences are the daily price moves, and they fluctuate around a small constant with no trend.

We compute them and read off the order of integration

The setup

- Eight daily prices trending 100 to 113.
- Clearly non-stationary in level.
- Candidate for one difference.

What we'll do by hand

- Take first differences.
- Check they're stationary.
- Confirm the order d is one.

Differencing, Precisely

Turn a level into a change

$$\Delta y_t = y_t - y_{t-1} \quad (\text{the operation at the heart of ARIMA: turn a level into a change})$$

What it does

- Each value becomes its change from before.
- A linear trend becomes a constant.
- A stochastic trend becomes stationary noise.

In finance

- Differencing a price gives the price change.
- Or, in logs, the return.
- Prices are $I(1)$; returns are $I(0)$.

Take the Differences

The level trends, the changes don't

price	100	103	101	105	108	106	110	
difference	-	+3	-2	+4	+3	-2	+4	+

The prices climb from 100 to 113, but their differences - +3, -2, +4, and so on - hover around a small positive drift of about 1.9 with no trend of their own. One difference has made the series stationary.

The Order of Integration

And difference versus detrend

prices 100, 103, 101, 105, ... $\Rightarrow \Delta y = 3, -2, 4, 3, -2, 4, 3$ (drift 1.86, no trend $\Rightarrow d=1$)

Reading d

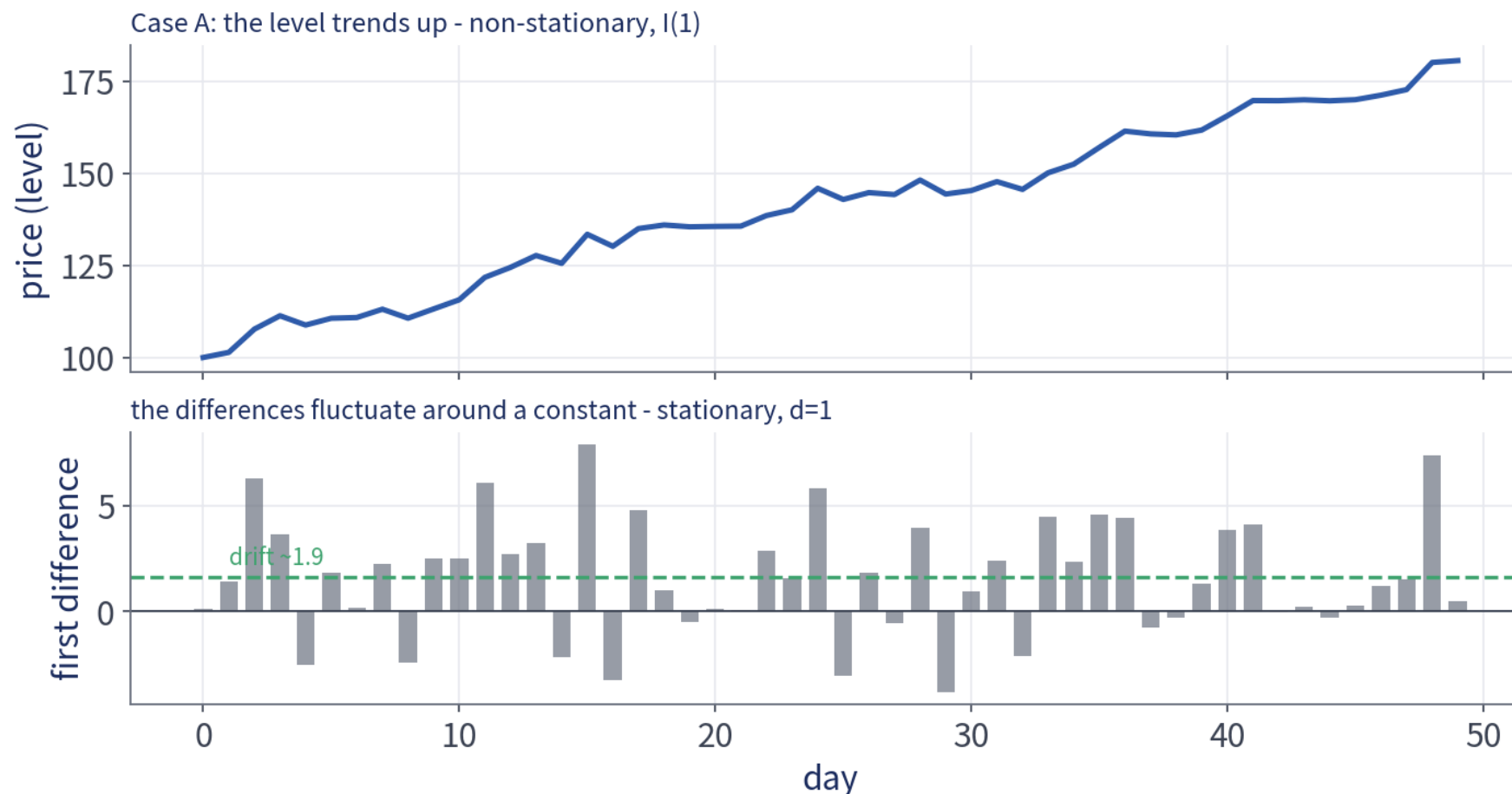
- The level fails a stationarity test.
- The differences pass it.
- So d equals one - written ARIMA(p,1,q).

Difference vs detrend

- Difference for a stochastic trend.
- Detrend - regress on time - for a deterministic one.
- ARIMA is for the stochastic kind.

The Trend Vanishes in the Differences

The top panel shows the price wandering steadily upward - non-stationary. The bottom panel shows its first differences, fluctuating around a constant drift with no trend at all. One differencing operation has stripped away the stochastic trend, leaving a stationary series to model



Why Difference at All

Prices versus returns

The problem with levels

- Prices trend, so ARMA can't model them.
- Their mean and variance drift over time.
- Regressions on them can be spurious.

The differencing fix

- The changes are stationary - ARMA applies.
- In finance, those changes are returns.
- One difference unlocks the whole toolkit.

Identify and Forecast

Model the change, then integrate back

Once differenced, the series is just ARMA - so we identify the order and forecast. The differences here look like a constant drift, making the price a random walk with drift: ARIMA(0,1,0) with a trend. We forecast the differences, then integrate them back into a level forecast that continues the trend, with bands that widen as the

The identification

- Difference once - d is one.
- The differences are a constant drift.
- So the model is a random walk with drift.

What we'll do by hand

- Forecast the drift forward.
- Cumulate it onto the last price.
- Widen the bands with the horizon.

ARIMA and Its Forecast

Difference, model, integrate back

$\text{ARIMA}(p, d, q)$: difference d times, then model as $\text{ARMA}(p, q)$; random walk = $\text{ARIMA}(0, 1, 0)$

The three integers

- p autoregressive, d differences, q moving-average.
- Difference d times to reach stationarity.
- Then model the rest as ARMA.

The forecast

- Forecast the differenced series first.
- Then cumulate to recover the level.
- The trend carries forward.

Integrate the Forecast Back

Add the drift, step by step

$$\hat{y}_{T+h} = y_T + \sum_{i=1}^h \widehat{\Delta y} = y_T + h\hat{\mu} : \quad 113 + 1(1.86) = \mathbf{114.9}, \quad 113 + 2(1.86) = \mathbf{116.7}$$

The mechanics

- Start from the last price, 113.
- Add the drift of 1.86 each step.
- That cumulates the differenced forecast.

The path

- 114.9, then 116.7, then 118.6.
- A straight-line continuation.
- The trend persists, it doesn't revert.

The Widening Bands

Uncertainty grows with the horizon

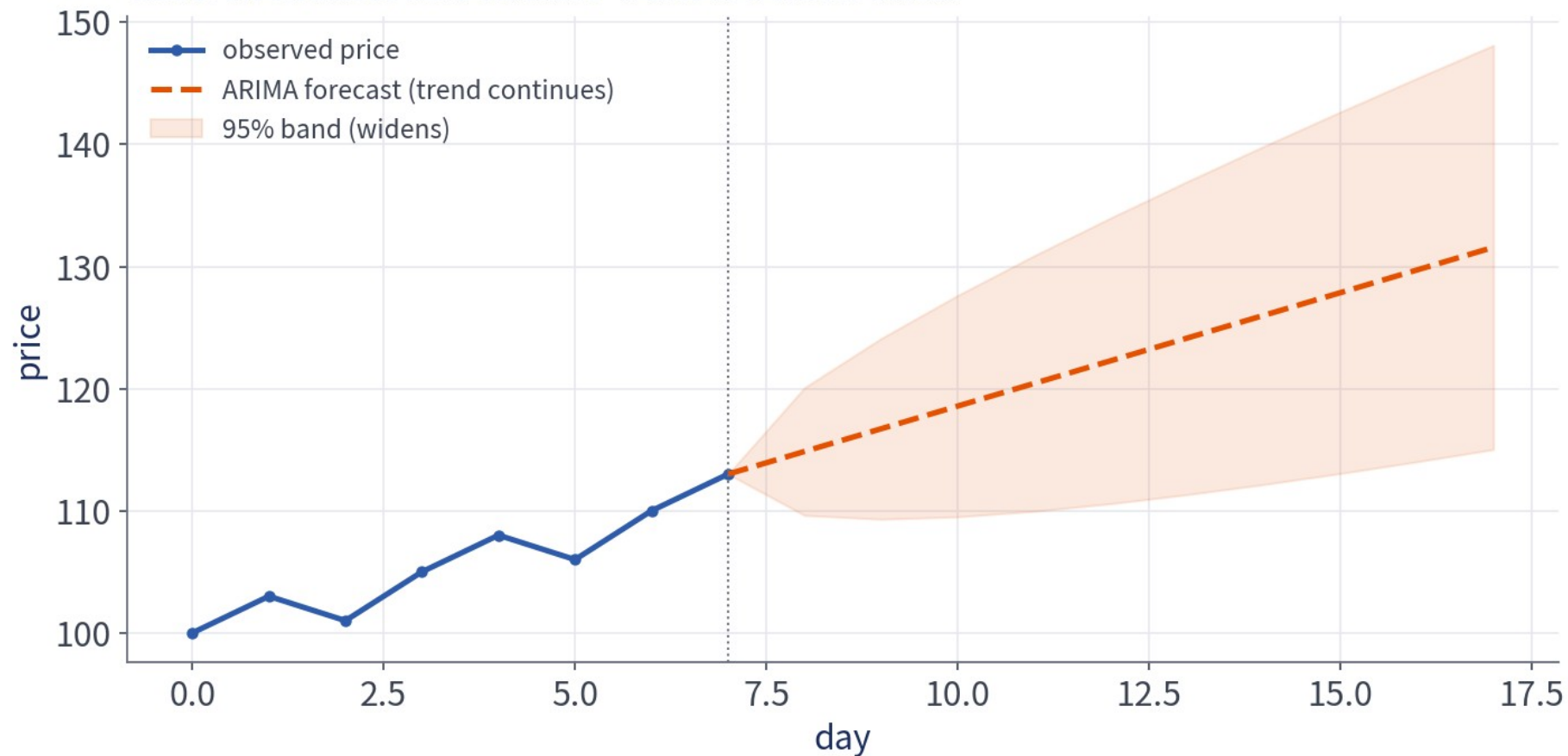
95% band = $\pm 1.96 \sigma \sqrt{h}$: ± 5.2 ($h=1$) $\rightarrow$ ± 9.1 ($h=3$) (trend continues, uncertainty grows)

Because the forecast integrates independent differences, its variance grows with the horizon - the band widens with the square root of h , from about plus-or-minus 5.2 at one step to 9.1 at three. The further out, the less certain the level.

The Trend Carries Forward

The observed prices climb, and the ARIMA forecast extends that climb in a straight line - the stochastic trend continues rather than reverting to any mean. The shaded band fans out with the horizon, because each step adds another independent uncertain difference to the running total

Case B: the stochastic trend continues - it does not revert to a mean



Trends Continue, Not Revert

The signature of a stochastic trend

What ARIMA forecasts

- A trend that continues indefinitely.
- Uncertainty that grows without bound.
- The behavior of a random walk with drift.

What a stationary model would do

- Revert to a fixed long-run mean.
- Bands that settle to a constant width.
- The wrong picture for a trending price.

The ARIMA Workflow

Difference, check model, integrate

The routine is four steps. Difference the series until it's stationary, using tests to fix the order d . Model the stationary differences as ARMA, identifying p and q . Forecast those differences. Then integrate back to recover the level, so the trend continues. One operation - differencing - bridges a trending price and the entire

The differencing (Case A)

- Prices 100 to 113, differences around 1.9.
- Stationary after one difference.
- Order d is one.

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The forecast (Case B)

- Drift forward: 114.9, 116.7, 118.6.
- Bands widen with the square root of h .
- The trend continues.

Reproduce the Differencing and the Forecast

Direct Claude Code to check your arithmetic

GOAL Difference a price series, confirm the order, then forecast by integrating back.

[INPUT] Prompt · Claude Code

Write and run `part1_practice.py` that:

CASE A - differencing:

- * for prices `[100,103,101,105,108,106,110,113]`, print the first differences and their mean;
- * run `adf` on the level and the differences.

CASE B - forecast:

- * treat it as a random walk with drift;
- * print the 1-, 2-, 3-step level forecasts;
- * print the 95% bands (widening with `sqrt(h)`).

Then plot the level and differences, and the forecast with its band.

Check against the slides

- differences +3,-2,+4,...; drift 1.86.
- level non-stationary, diffs stationary.
- forecast 114.9, 116.7, 118.6.
- bands +/-5.2 -> +/-9.1.

Deliverable: `part1_practice.py` whose printed numbers match the differences and the forecast path.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 What does the 'I' in ARIMA stand for?

Integration - the series is differenced d times

Q2 How do you pick the order d ?

Difference until stationarity tests pass

Q3 How does an ARIMA level forecast behave?

The trend continues; it doesn't revert to a mean

All three come from the differencing, the order of integration, and the integrated forecast.

Three Pitfalls to Avoid

Where ARIMA trips people

Common mistakes

- "Difference until it looks flat, just in case."
- "Detrending and differencing are the same."
- "An ARIMA forecast reverts to a mean."

The reality

- Over-differencing adds noise - test for the right d .
- Difference stochastic trends, detrend deterministic ones.
- The stochastic trend continues, with widening bands.

Two Calculations to Remember

Part 1 applications in one screen

1

One difference makes it stationary

The prices climbed from 100 to 113, but their first differences - +3, -2, +4, and so on - fluctuated around a drift of 1.86 with no trend. The level was non-stationary; the differences were stationary; the order of integration is one.

2

Integrate the forecast back

Treating the price as a random walk with drift, the level forecast added 1.86 each step - 114.9, 116.7, 118.6 - continuing the trend rather than reverting. The 95% band widened with the square root of the horizon, from 5.2 to 9.1.

3

Differencing bridges trend and toolkit

Difference until stationary, model the changes as ARMA, forecast, then integrate back. That single operation lets the full stationary machinery handle a trending price.

Next — Part 2 practice: seasonality and SARIMA, differencing away the yearly and weekly cycles too.